

CliniCause: Constructing Reusable Causal-Analysis Datasets from Irregular ICU Records

Anonymous submission

Abstract

Reusable causal analysis of heterogeneous, irregular intensive-care records requires explicit constructs, cohort boundaries, adjustment assumptions, and provenance that prediction-ready sequences do not provide. We present ClinCause, two validated causal-analysis datasets built from MIMIC-III and PhysioNet 2012. The resources contain 26,845 and 7,993 analysis records with nine and ten admitted proxy-state exposures, respectively, linked to in-hospital mortality, observed covariates, dataset-specific causal-graph metadata, and provenance. An evidence-tracked pipeline combines deterministic proxy rules with annotations from four irregular-time-series models and enforces schema, cohort, artifact, and provenance contracts; a design-time LLM proposes candidates but does not process patient records or estimate effects. On archived predictive point metrics, STraTS led all four MIMIC-III metrics and GRU-D led all four PhysioNet metrics. Across 19 prespecified dataset-exposure comparisons, the two double-machine-learning estimators agreed in direction for 19/19, while CausalPFN joined them in 18/19; PhysioNet shock was the sole all-estimator exception. These observational, proxy-based results do not establish clinical effects or causal identification. They show that ClinCause supports reusable, inspectable analysis and that estimator triangulation can expose both stable directional patterns and informative disagreements across heterogeneous ICU resources.

1 Introduction

Intensive-care records combine measurements, care processes, and outcomes observed at unequal frequencies and times. Their missingness can reflect both patient state and clinical workflow (Sun et al. 2026; Lipton, Kale, and Wetzel 2016). Causal analysis therefore requires more than a prediction-ready sequence: it needs explicit analysis units, exposures, outcomes, covariates, cohort boundaries, adjustment assumptions, and provenance. Reconstructing these interfaces study by study makes silent changes to constructs or cohorts hard to detect. Identifier conventions, measurement availability, and preprocessing can alter which records reach analysis even when studies begin from the same database. A reusable resource must therefore preserve both variables

and the lineage and assumptions that give them analytical meaning.

ICU databases and benchmarks provide source records and prediction tasks (Johnson et al. 2016; Silva et al. 2012; Harutyunyan et al. 2019); irregular-time-series models represent sparse events or measurement gaps (Tipirneni and Reddy 2022; Che et al. 2018); electronic phenotyping and weak supervision turn clinical evidence or heuristics into analytical labels (Banda et al. 2018; Ratner et al. 2016); and causal machine learning estimates adjusted, potentially heterogeneous contrasts (Chernozhukov et al. 2018; Wager and Athey 2018). What remains difficult is their resource-level integration: carrying replaceable proxy constructs from heterogeneous irregular records through predictive annotation into graph-documented, estimator-ready datasets with validation and provenance. Without that interface, targets, exposures, adjustment variables, and populations may each be defensible yet difficult to reuse or compare as one traceable scientific object. The challenge is not simply to concatenate components, but to make every handoff explicit enough that a construct, cohort, or adjustment decision can be changed and audited without silently redefining the rest of the analysis.

CliniCause constructs two such resources from MIMIC-III and PhysioNet 2012 under a common interface while preserving dataset-specific semantics. Rules define project-selected proxy states; four temporal models add predictive annotations; and each resource joins aggregate proxy exposures to in-hospital mortality, observed covariates, causal-graph adjustment metadata, cohort identity, and provenance. A design-time LLM proposed candidates that were project-selected and encoded as deterministic source before patient-level execution. The datasets and estimates remain separate. Here, *validated* denotes construction and analytical integrity—aligned schemas, cohorts, artifacts, provenance, and analytical use—not clinical construct validation or causal identification.

The resources contain 26,845 MIMIC-III records with nine admitted exposures and 7,993 PhysioNet records with ten. STraTS led all four archived MIMIC-III point metrics, whereas GRU-D led all four PhysioNet metrics. CausalForestDML and LinearDML agreed in direction for 19/19 prespecified comparisons, CausalPFN joined them in 18/19, and 55/57 original-versus-downsampled estimator comparisons preserved direction.

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Our contributions are: (1) two reusable, validated causal-analysis datasets; (2) an evidence-tracked pipeline integrating deterministic proxies, temporal prediction, causal-graph metadata, validation, provenance, and bounded design-time LLM assistance; (3) predictive and three-estimator characterization, including 19/19 DML and 18/19 all-estimator directional agreement; and (4) interfaces for replacing proxies, temporal models, graphs, adjustment sets, estimators, and robustness analyses without repeating upstream integration. The hierarchy keeps the resources primary, their construction pipeline second, and empirical analyses as demonstrations of utility.

2 Related Work

MIMIC-III provides heterogeneous critical-care records, while PhysioNet 2012 and the MIMIC-III benchmark define prediction tasks (Johnson et al. 2016; Silva et al. 2012; Harutyunyan et al. 2019). GRU-D models informative missingness and STraTS represents sparse time-variable-value events (Che et al. 2018; Tipirneni and Reddy 2022); GRU and TCN supply recurrent and convolutional baselines (Cho et al. 2014; Bai, Kolter, and Koltun 2018). CliniCause uses these model families to annotate explicit proxies for downstream causal analysis. Unlike prediction benchmarks, the final object here includes exposure, adjustment, cohort, and provenance interfaces for estimator comparison. Thus temporal models are evaluated as annotation mechanisms inside a larger resource contract rather than as mortality-prediction endpoints.

Electronic phenotyping spans inspectable rules and learned models (Banda et al. 2018); data programming and Snorkel formalize heuristic labeling sources (Ratner et al. 2016, 2020). CliniCause uses no trained label model: it preserves dataset-specific deterministic definitions and applies a fixed majority vote over one rule and four model sources, keeping construct lineage explicit. This retains the inspectability of programmatic labeling while separating the fixed aggregation from methods that learn source accuracy or dependence. The resulting vote is consequently an implemented analytical construct, not an estimate of diagnostic truth or annotator consensus.

Double machine learning accommodates flexible nuisance estimation through orthogonal scores and cross-fitting (Chernozhukov et al. 2018); causal and generalized random forests model heterogeneous effects under their assumptions (Wager and Athey 2018; Athey, Tibshirani, and Wager 2019). CausalPFN instead amortizes effect estimation in a prior-data-fitted, in-context model trained on simulated data-generating processes (Meresht et al. 2025). These methods provide estimators, whereas CliniCause contributes the reusable exposure, outcome, covariate, cohort, graph, adjustment, and provenance interfaces on which they are compared. Estimator innovation alone does not supply a traceable clinical analysis table or preserve the construct decisions that produced it. CliniCause complements estimator research by holding these inputs and decisions in an inspectable, dataset-specific resource boundary.

Medical LLM studies show capability alongside reliability gaps, and causal-graph work treats LLM judgments as fallible

priors (Singhal et al. 2023; Darvariu, Hailes, and Musolesi 2024). CliniCause therefore restricts the LLM to inspectable design proposals and leaves runtime authority to deterministic source. Its contribution is the traceable integration of these otherwise separate strands into two reusable causal-analysis resources; the LLM is a bounded proposal mechanism rather than a clinical or estimation authority. This division preserves the benefit of structured elicitation while leaving both the accepted proposal and its deterministic implementation open to review.

3 Constructing the CliniCause Datasets

CliniCause constructs two reusable, validated causal-analysis datasets from MIMIC-III and PhysioNet 2012. Each connects irregular observations to proxy exposures, in-hospital mortality, observed covariates, and inspectable graph metadata under a shared interface that retains source-specific semantics (Figure 1; Table 1).

3.1 Source Cohorts and Analysis Units

The resources originate from MIMIC-III and the PhysioNet/Computing in Cardiology Challenge 2012 collection (Johnson et al. 2016; Silva et al. 2012). An analysis unit is a source time-series record or ICU stay, not necessarily a unique person. Source identifiers map to a canonical `ts_id`; irregular events retain time, variable, and value, while a record table supplies mortality and available baseline covariates. Linked long-event and record-level views support temporal modeling and causal analysis without flattening irregular events into the estimator table prematurely. Split-aware predictive artifacts and the complete admitted causal cohort remain distinct and are aligned by validation contracts rather than treated as interchangeable files.

The admitted populations contain 26,845 MIMIC-III and 7,993 PhysioNet records. Their covariates, measurement processes, proxy ontologies, and graphs remain dataset specific, so records and estimates are never pooled and similarly named constructs are not assumed equivalent. The common interface standardizes analytical roles—identifier, exposure, outcome, covariate, adjustment, and provenance—rather than forcing a common clinical vocabulary onto the two sources. Portability can therefore be evaluated without erasing source-specific measurement and construct boundaries.

3.2 Proxy-State Design and Deterministic Construction

Before patient-level execution, a structured LLM protocol proposed proxy ontologies, binary-rule families, missingness considerations, and dataset-specific graphs. Project-selected proposals were encoded in deterministic tagger and graph source. That source, not prompt text, governs runtime; the LLM neither processes patient records nor assigns exposures or estimates effects. This boundary makes proposals inspectable while ensuring that patient-level execution depends on versioned deterministic artifacts.

Dataset-specific rules map measurements and care-process indicators to binary proxy fields while preserv-

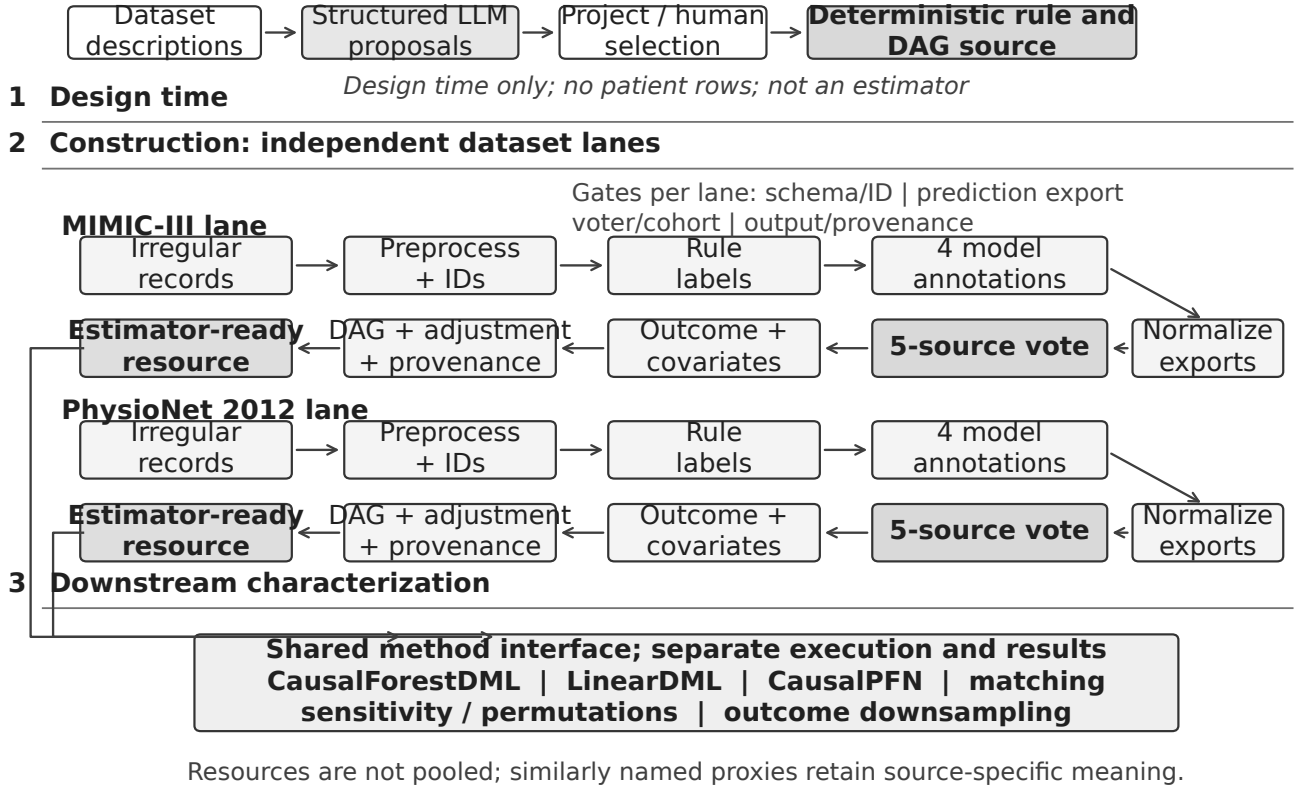


Figure 1: CliniCause construction. Separate dataset lanes combine deterministic proxy rules and four normalized model annotations, then package five-source aggregates with outcomes, covariates, graph metadata, and provenance. Validation gates protect each handoff. The LLM proposes candidates only at design time; selected deterministic source governs runtime.

ing source differences (e.g., combined MIMIC-III hepatic/coagulation evidence versus separate PhysioNet constructs). Missing numeric measurements provide no positive rule evidence, making availability part of construct provenance. These are transparent analytical proxies, not chart-adjudicated diagnoses. Their explicit columns provide a replaceable interface: reviewers can revise a rule without silently changing the outcome, covariates, or cohort contract. Missing measurements cannot create positive rule evidence by being silently interpreted as normal; the observed input envelope remains part of each proxy’s meaning.

3.3 Predictive Annotations from Irregular Time Series

Each predictive task estimates its dataset’s rule-derived proxy vector. STraTS represents sparse observation times, values, and variables; GRU and TCN process temporal grids; and GRU-D models decay from missingness and time gaps (Tipirneni and Reddy 2022; Cho et al. 2014; Bai, Kolter, and Koltun 2018; Che et al. 2018). Static covariates join each temporal representation. Value, observation-mask, and elapsed-time channels preserve different aspects of irregular sampling across architectures. Targets are proxy states, not mortality.

Normalized exports pair target probabilities with binary

predictions at the implemented 0.5 threshold. Archived causal runs combine one rule-derived source and four aligned model sources by deterministic majority vote. This aggregate is an algorithmic proxy-exposure layer, not a learned label model or expert consensus. Predictive metrics characterize recovery of rule targets, not their clinical validity. Probability and binary fields remain paired in normalized exports, while aggregation consumes only aligned binary voters; prediction evaluation therefore remains distinct from exposure assembly. Complete voter sets are required so that a missing model export cannot silently change the aggregation denominator or admitted cohort.

3.4 Causal-Analysis Schema and DAG Metadata

Estimator-ready rows align canonical identifiers, aggregate proxy fields, in-hospital mortality, and source-specific covariates. Linked metadata records dataset, sampling condition, graph, run, and artifact provenance. The analysis admits nine non-chronic MIMIC-III and ten non-chronic PhysioNet exposures; chronic-burden fields remain baseline constructs rather than analyzed exposures. Exposure, outcome, adjustment, and provenance roles are explicit even when their fields reside across linked tables and metadata.

For each exposure, a source-coded graph maps project

Dataset	Records	Exposures	Outcome	Predictive annotations	Causal-analysis estimators	
MIMIC-III	26,845	9	In-hospital mortality	1 rule source + 4 models	CF-DML, CausalPFN	LinearDML,
PhysioNet 2012	7,993	10	In-hospital mortality	1 rule source + 4 models	CF-DML, CausalPFN	LinearDML,

Table 1: Primary original-sampling resources. Counts are analysis records and admitted non-chronic proxy exposures, not raw cohorts; CF-DML denotes CausalForestDML.

assumptions to observed columns and an exposure-specific adjustment set. The graphs are neither learned nor clinically validated ground truth, but encoding them makes adjustment choices inspectable and replaceable alongside proxies and estimators without rebuilding source-data integration. Alternative graphs can consequently be compared against the same aligned resource instead of being conflated with a new preprocessing population. Graph or estimator changes thus remain visible analytical choices rather than hidden changes to the data resource.

3.5 Dataset Validation and Provenance

Structural validation checks files, column order, identifiers, values, and output completeness. Cohort contracts require identifier preservation and exact equality at handoffs, rejecting duplicates, conflicts, missing or extra records, and silent shrinkage. Unexpected columns, invalid voter/probability fields, binary inconsistencies, and incomplete artifact sets fail explicitly. Together, these checks protect one analytical population from deterministic labels through predictive annotations and estimator-ready assembly.

Validation also spans artifact identity and execution boundaries. Prediction, checkpoint, and voter sidecars bind model and input roles; run receipts distinguish planned from completed stages; upstream fingerprints prevent reuse after source mutation; and dataset-isolated paths prevent cross-source artifact substitution. A handoff is validated only when its required evidence is present, rather than inferred from a file name or partial directory.

Artifact validation binds outputs to supported metadata, hashes, schemas, cohorts, configurations, and seeds. Dataset-isolated run paths, manifests, receipts, and sidecars record roles and outputs; stale or mutation-mismatched artifacts fail reuse checks. Deterministic transformations and derived seeds avoid dependence on ambient order or process state. These contracts concern the current resource interface and do not recover historical run provenance by themselves.

The checked archive establishes valid export schemas, complete primary aggregates, manifests, checksums, and archived analysis-stage records, but exact producing revisions, configurations, and checkpoint-to-export links remain incomplete. The repaired current code contains stricter source/test contracts; it is not claimed as the historical producer, nor do we claim current test execution. Keeping these evidence layers separate avoids turning checked output integrity into an unsupported exact-rerun claim.

Both resources supported predictive characterization, three effect estimators, graph-guided adjustment, matching,

and robustness diagnostics. Validation thus establishes structural integrity, cohort consistency, provenance, and analytical reuse; clinical construct validity and causal identification remain separate questions. This scoped definition permits strong resource validation without promoting proxy states to diagnoses or adjusted contrasts to proven effects. It also distinguishes resource reuse from full historical reproducibility: the former is demonstrated by explicit contracts and completed analyses, while the latter still requires missing producing lineage and a verified rerun.

4 Empirical Evaluation

The evaluation asks three linked questions. First, can irregular-time-series models recover the constructed rule targets, and does their behavior differ by dataset? Second, are the principal exposure–outcome directions stable across distinct estimator forms? Third, do matching and population perturbations expose support or sign sensitivity? These questions characterize the resources and their analytical interfaces; they do not test a clinical intervention.

4.1 Prediction Tasks and Metrics

For each dataset, STraTS, GRU, GRU-D, and TCN predict its deterministic proxy vector. We compare one selected archived held-out test output per model–dataset combination, without pooling targets or scores. Because repeated-run uncertainty is unavailable, these point metrics characterize prediction of rule-derived constructs, not statistical superiority or clinical validity. Target sets follow the source-specific schemas, so the evaluation asks whether each constructed proxy vector can be recovered from its own irregular ICU representation. The comparison includes all eight archived model–dataset outputs selected for the main analysis rather than choosing models by a single favorable metric.

We report sample-mean multi-label binary cross-entropy loss and macro AUROC, AUPRC, and minRP. Per target, minRP maximizes the smaller of precision and recall over thresholds. The three discrimination scores are averaged after excluding single-class targets; loss follows the archived sample-mean evaluator output. No confidence interval or significance test is inferred from the selected runs.

4.2 Effect Estimators and Cross-Estimator Evaluation

Each admitted aggregate proxy is analyzed separately as a binary exposure, with in-hospital mortality as outcome and observed adjustment variables selected by the project graph. Every prespecified non-chronic exposure is retained. The

original populations are primary, and datasets are analyzed without pooling. This complete-set policy prevents favorable post-hoc exposure selection. The graphs encode the project’s observed-adjustment assumptions; they do not remove requirements for consistency, overlap, well-defined interventions, or absence of unmeasured confounding.

CausalForestDML is the primary nonlinear estimator, combining DML adjustment with a causal-forest final stage (Chernozhukov et al. 2018; Wager and Athey 2018; Athey, Tibshirani, and Wager 2019; Oprescu et al. 2019) to obtain record-level fitted conditional effects. LinearDML retains the DML nuisance framework with a linear final effect model, isolating a change in final-stage form. CausalPFN (Meresht et al. 2025) provides a distinct complementary paradigm over the same prespecified comparisons; its archived estimator-specific sensitivity and permutation coverage is less complete than for DML.

We summarize fitted conditional estimates by their *mean model-estimated CATE over the analyzed sample*, not as a risk ratio, odds ratio, or unqualified population ATE. We compare its sign between the DML estimators and across all three estimators, retaining disagreements. Directional agreement neither requires equal magnitudes nor proves causal identification; it tests whether the principal sign pattern depends on one estimator family. All interpretations are therefore observational and proxy based, and none is a treatment recommendation. Magnitude remains visible in Figure 2 so sign concordance does not collapse substantively different fitted summaries into equivalence.

4.3 Matching and Robustness Diagnostics

Matching is a descriptive support check: binary adjustment variables and median-binarized numeric variables enter greedy one-to-one Hamming matching. The procedure begins with exact binary matches and increases allowed distance within its configured pair-count and match-rate criteria. We report a *descriptive matched-pair outcome difference* with availability, distance, and support warnings; no usable representation produces an explicit failure, not a zero effect or an imputed estimate.

Archived DML sensitivity records retain native, recomputed/reconstructed, partial, failed, and unavailable provenance classes. Treatment and outcome permutations are disruption checks rather than formal randomization tests or identification proofs. Equivalent stages were not archived for CausalPFN, which marks diagnostic-coverage asymmetry rather than exclusion from triangulation. For DML, recorded sensitivity provenance is preserved because estimator-native, later reconstructed, partial, and failed evidence are not interchangeable forms of validation.

Outcome downsampling retains mortality-positive records and samples negatives, changing both population and prevalence. It is a robustness perturbation, not a co-primary analysis, and its magnitudes are not interchangeable with original-population estimates. Applying the shared sequence separately to both resources tests workflow portability while preserving dataset-specific definitions, cohorts, and causal assumptions. The matching, disruption, and downsampling analyses are supporting diagnostics; no single check up-

grades adjusted observational contrasts into identified intervention effects.

5 Results

The resources comprise 26,845 MIMIC-III records with nine exposures and 7,993 PhysioNet records with ten. All analyses remain dataset specific and unpooled.

5.1 Predictive Characterization

STraTS led all four archived MIMIC-III point metrics, whereas GRU-D led all four PhysioNet metrics (Table 2). This dataset-dependent leadership does not imply statistical superiority. STraTS reached 0.905 AUROC and 0.869 AUPRC on MIMIC-III; GRU-D reached 0.918 and 0.905 on PhysioNet. Each also led loss and minRP within its dataset, so leadership was consistent across the four archived metrics against rule-derived targets. The change in leader reinforces the resource design choice to retain source-specific measurement and missingness semantics rather than select one temporal representation globally.

5.2 Effect Patterns and Estimator Agreement

All nine MIMIC-III CausalForestDML mean summaries were positive, as were nine of ten PhysioNet summaries. MIMIC-III was led by cardiac strain (0.220) and inflammation/sepsis (0.161); PhysioNet was led by renal dysfunction (0.120), cardiac injury/strain (0.112), and global severity (0.108), while shock was negative (−0.014). These are mean model-estimated CATE over the analyzed samples under the implemented proxy and adjustment design; magnitudes and construct orderings remained dataset specific (Figure 2).

CausalForestDML and LinearDML agreed in direction for all 19 prespecified comparisons. CausalPFN joined their direction in 18/19, providing triangulation from a distinct modeling paradigm while retaining its smaller archived diagnostic envelope. Complete DML concordance shows that the sign pattern was not unique to a linear or forest final stage, while CausalPFN’s broad agreement extends the observation beyond that shared DML framework. Neither result implies equal magnitudes or identifies a causal effect.

PhysioNet shock was the sole exception: CausalForestDML was −0.014, LinearDML −0.027, and CausalPFN 0.004; its descriptive matched-pair difference was 0.010. The disagreement motivates targeted analysis of the proxy, support, adjustment assumptions, and estimator behavior rather than estimator voting. Retaining this near-zero exception is important because a direction-only summary would otherwise conceal exactly where model choice changes the conclusion.

5.3 Robustness and Cross-Dataset Findings

Matching succeeded for 15/19 comparisons (eight MIMIC-III and seven PhysioNet); four failures lacked usable binary columns and are not zero effects. Of the 15 summaries, 14 agreed with the primary direction; PhysioNet shock did not. The availability split also distinguishes empirical support limitations from estimator disagreement on completed analyses. Matching agreement is descriptive corroboration only:

Dataset	Model	Test loss↓	Test AUROC↑	Test AUPRC↑	Test minRP↑
MIMIC-III	STraTS	0.348	0.905	0.869	0.795
	GRU	0.386	0.881	0.840	0.770
	GRU-D	0.404	0.884	0.841	0.773
	TCN	0.414	0.867	0.823	0.758
PhysioNet 2012	STraTS	0.341	0.915	0.877	0.818
	GRU	0.397	0.915	0.897	0.831
	GRU-D	0.331	0.918	0.905	0.835
	TCN	0.477	0.899	0.875	0.811

Table 2: Selected archived held-out point metrics against rule-derived proxy targets. Bold marks each dataset’s best value; no significance is implied.

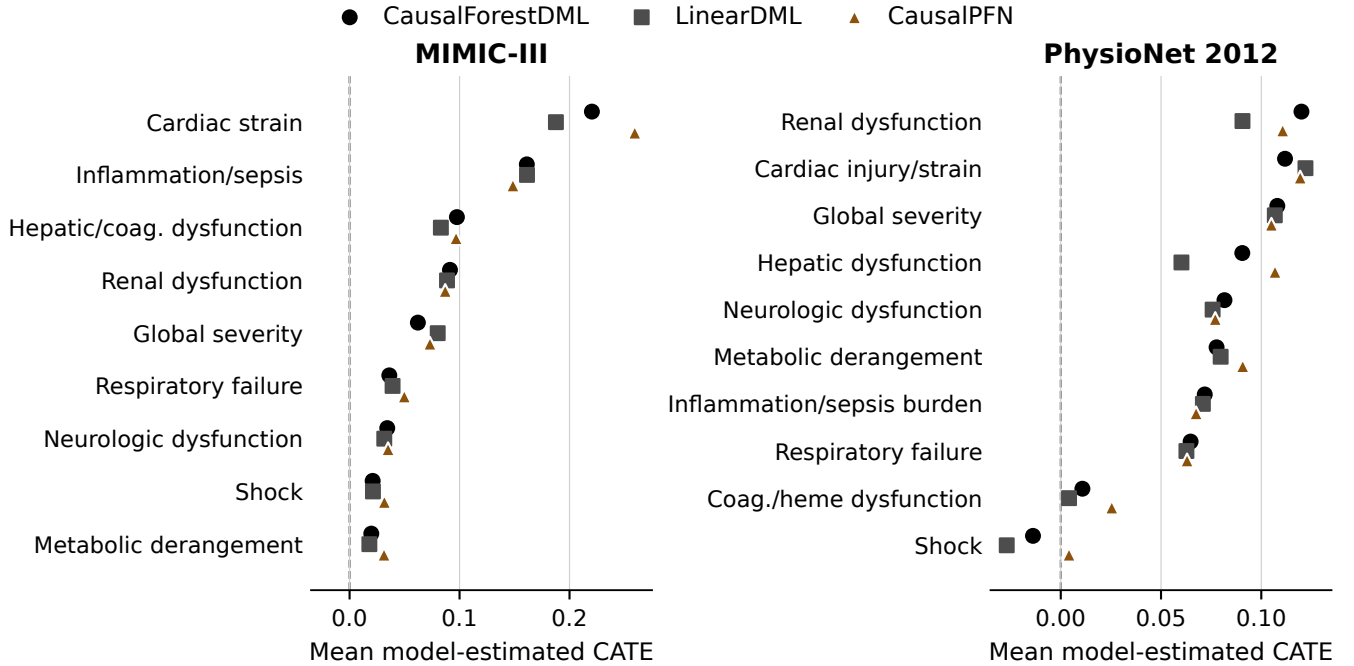


Figure 2: All 19 original-cohort comparisons, ordered by CausalForestDML. Markers are mean model-estimated CATE over each analyzed sample. DML directions agree in 19/19 and all three in 18/19; PhysioNet shock is the exception.

it neither repairs failed overlap nor supplies an independent identified causal effect.

Direction persisted in 55/57 original-versus-downsampled comparisons. The two changes were PhysioNet LinearDML coagulation/hematologic dysfunction and PhysioNet CausalIPFN shock; magnitudes do not transport to the original population. DML sensitivity and permutation evidence has nonuniform provenance and coverage, with no equivalent archived CausalIPFN stages; these diagnostics remain supporting evidence rather than independent identification tests. Together, differing predictive leaders and effect patterns alongside broad estimator agreement show workflow portability, not pooled clinical replication.

6 Discussion and Limitations

6.1 Resource Utility and Cross-Dataset Portability

CliniCause contributes reusable resources rather than only effect tables. Explicit exposure, outcome, covariate, graph, cohort, and provenance interfaces support comparisons of estimators and temporal representations without repeating every integration stage. Deterministic definitions and lineage also make proxy, graph, adjustment, cohort, and support alternatives directly auditable. A study can replace one layer while retaining the remaining analytical contract, making the present estimates starting points for controlled follow-up rather than fixed endpoints. Concretely, the same tables can support estimator benchmarking; the event-to-proxy interface can test new temporal representations; and preserved construct and graph metadata can support clinician-led proxy

or adjustment revision. Each use keeps cohort identity visible, helping distinguish a method change from an unnoticed population change. Because resource validation is separated from clinical and causal validation, follow-up studies can strengthen one evidence layer without discarding the audited interfaces already established in the others.

The validation contracts are scientifically consequential rather than only software safeguards. Exact cohort equality prevents a join from silently changing the population summarized by an estimator. Schema and voter checks prevent missing annotations from redefining an exposure. Graph and adjustment metadata expose assumptions that would otherwise be implicit, while hashes and receipts distinguish a known artifact from a similarly named replacement. Together these controls make differences between analyses attributable to declared choices instead of undetected pipeline drift. They do not establish that a proxy or graph is correct, but they make those scientific questions answerable on a stable resource.

The same sequence operated across heterogeneous ICU sources although predictive leadership, magnitudes, and rankings differed. Portability therefore resides in the workflow and interface, not construct equivalence: variables, measurement processes, ontologies, graphs, and analyses remain dataset specific and unpooled. This separation exposes contextual sensitivity instead of hiding it behind a pooled replication claim. It also lets future work distinguish success of a common interface from equivalence of source semantics. Operation on two sources demonstrates workflow portability, not external clinical generalization of any proxy or effect estimate. The source differences also illustrate why a common interface is preferable to a forced common ontology. MIMIC-III and PhysioNet vary in available variables, sampling processes, and whether hepatic and coagulation evidence form one proxy or two. Preserving those differences allows researchers to compare methods while tracing disagreement to representation, construct, adjustment, support, or estimator choices. A pooled table would make that attribution harder and could give similarly named columns a false clinical equivalence.

6.2 Estimator Triangulation and CausalPFN

The 19/19 DML and 18/19 all-estimator sign agreements show that the prevailing pattern extends across final-stage forms and to CausalPFN's distinct paradigm. PhysioNet shock usefully exposes model and support sensitivity for targeted study. Directional agreement does not establish equal magnitudes, uncertainty, or causal identification; CausalPFN's smaller archived diagnostic envelope also motivates estimator-specific follow-up. Broad concordance plus one visible exception is more informative than an aggregate vote: it supports further CausalPFN evaluation while identifying a concrete case for proxy, adjustment, overlap, and estimator analysis. For future estimator studies, the complete 19-comparison matrix supplies both a broad concordance target and a prespecified stress case. Repeated runs, estimator-specific uncertainty, and harmonized diagnostics should determine whether the shock disagreement is sampling variation, weak support, construct sensitivity, or a genuine modeling difference.

6.3 LLM-Assisted Design and Future Research

The LLM layer is a design aid for inspectable proxy, missingness, and graph proposals; project selection and deterministic source retain authority. Future work can use the resources to test clinician-reviewed proxies, expert/LLM or design-layer alternatives, graphs, repeated-run uncertainty, richer CausalPFN diagnostics, and external datasets. Because proposals remain separate from patient execution, such comparisons need not grant the LLM runtime clinical authority. Prompt sensitivity and proposal-to-source traceability should also be evaluated so that any design benefit is distinguishable from downstream human selection and deterministic implementation.

6.4 Limitations

Proxy states are analytical constructs, not chart-adjudicated phenotypes, but their definitions and lineage permit clinician-led replacement and validation. Project graphs do not prove intervention validity; observational estimates remain vulnerable to unmeasured confounding, identification, and support failures and are not treatment recommendations. Alternative graphs, adjustment sets, and stronger support analyses are necessary before stronger interpretation. Repeated-run uncertainty and diagnostic coverage are incomplete, especially for CausalPFN. Exact producing revisions, configurations, and checkpoint-to-export lineage also remain incomplete; current runtime and anonymous release are separately gated. Accordingly, archived point-metric leaders are not statistical rankings, the agreement counts are not uncertainty estimates, and the present artifacts do not support an exact clean-checkout reproduction claim. Restricted source-data access, absent external clinical validation, and no isolated LLM ablation further limit generalization. The resource interfaces make these boundaries explicit and testable rather than burying them in undocumented pipeline choices. Replaceability facilitates those studies but is not itself evidence that a replacement, proxy, or graph is clinically valid.

7 Conclusion

CliniCause constructs reusable, estimator-ready resources from MIMIC-III and PhysioNet 2012 with validated structural, cohort, provenance, and analytical interfaces. Broad DML/CausalPFN directional agreement demonstrates bounded estimator triangulation: the DML estimators agreed in 19/19 directions and CausalPFN joined them in 18/19, with PhysioNet shock retaining the informative exception. Archived predictive leadership also differed by dataset, reinforcing the need to preserve source semantics. These transparent resources support tests of proxy definitions, temporal representations, graphs, cohorts, and estimators while preserving the observational and proxy-based boundary of the findings. This foundation is intended for transparent comparison and revision, not patient-level deployment or treatment selection. Its value lies in making the next proxy, graph, representation, or estimator comparison easier to specify, audit, and interpret.

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